F78AA Actuarial and Financial Mathematics

Chapter 1: Introduction

What is Financial Mathematics?

- It is a branch of mathematics dealing with **financial contracts** and **financial markets**.
- Can you think of any examples of financial contracts?

E.g. 1.2.1: Bank account paying interest

- You open a savings account with Bank XYZ with a deposit of £1,000. At the end of one year the account will be closed and you will receive £1,050 from the bank.
- Draw the timeline.
- How would you show the difference in the cashflows from the point of view of the investor and the bank?

E.g. 1.2.1: Bank account paying interest (Solution)

You and the bank have exchanged two sets of cashflows; one consisting of your deposit of £1,000, and the other consisting of the payment to you of £1,050 in one year's time.

You (investor or lender)	Bank XYZ (borrower)

E.g. 1.1: Bank account paying interest (Solution 2)

If the Bank does monthly accounts management then the timeline would be:

E.g. 1.1: Bank account paying interest (Solution 3)

The timeline may also convey the exact dates of the cashflows for e.g.

Note that the convention for the date we use in this module is dd/mm/yy.

E.g. 1.2.2: Mortgage (Home Loan)

- You take out a mortgage for £100,000 from a bank and repay this loan by making payments of £400 at the end of every month for 25 years.
- Draw the timeline.

You and the bank have exchanged two sets of cashflows; one consisting of £100,000 now and the other consisting of a stream of payment of £400 a month for 25 years.

E.g. 1.2.3: Fixed interest bond

- Suppose on 4/9/25 you buy £10,000 worth of a fixed interest investment that pays 4% interest per annum (p.a. OR *pa*). Interest each year is paid once a year on 4 September and the bond matures at par on 4/9/35 (i.e. the £10,000 is repaid to you on this date).
- Draw the timeline.

E.g. 1.2.4: Purchasing Share/Stock

- Suppose you buy £10,000 worth of shares in a quoted company on the London Stock Exchange. In return, you have the right to receive all the future dividends allocated to those shares, declared in future, until you decide to sell the shares, at which point you can sell them on the stock market for their then-current market price.
- Draw the timeline.

E.g. 1.2.5: Life Insurance contract

- You take out a whole of life insurance contract. This will pay out £100,000 on your death, whenever that might be. You pay a premium of £100 per month to the insurance company.
- Draw the timeline
- How could you determine the Present Value of this contract?

E.g. 1.2.6: House insurance contract

- You pay £500 to an insurance company, that will pay for any damage to your house and contents caused by fire in the next year.
- What is the problem with the timeline in this case?
- Can you evaluate the present value of the contract?

Cashflows might not be known in advance

As we have developed the **timelines** for the examples above, we have encountered the properties of cashflows that characterise a financial contract, namely:

- Direction
- Timing
- Amount
- Number of cashflows

Cashflows might not be known in advance

What information is missing from the contracts we have just considered?

- Savings account
- Mortgage
- Fixed Interest Bond
- Share
- Life insurance
- House insurance

Example	Direction	Timing	Amount	Number
Bank account				
Mortgage				
Fixed Interest Bond				
Share				
Life insurance				
House insurance				

Further uncertainty

- In example 1.2.1 the bank where you saved your money could go bust before it repays you. This is called **credit risk** or **counterparty risk**.
- In example 1.2.2 you might decide to repay your mortgage early, or you may die before the end of the term.
- In example 1.2.3 the company that issued the bond may go bankrupt.
- In example 1.2.4 the company may go bankrupt.

cont.

Further uncertainty

- In example 1.2.5 you might cancel your insurance contract or the insurance company might go bust before you make your claim.
- In example 1.2.6 the insurance company might not be able to pay your claim, perhaps due to catastrophic losses elsewhere.

This course does not deal with **uncertainty**. How we assess this uncertainty is dealt with in other courses.

REQUIRED READING

1.3: Quantitative Questions Arising From Financial Contracts (pages 14 and 15 of the Lecture Notes)

Next Chapter

Chapter 2: Interest and Present Value